

Algorithms 2

Linear Programming

Daniel Rosenberg & Michael Trushkin

Ariel University

1. Linear Programming

- **Example:**

- In a power grid, we have n power plants and m cities. We want to decide how much electricity each plant should produce to meet the demand at minimum cost.
- Let x_i be the amount of electricity produced by plant i .
- Let b_j be the demand at city j .
- Denote by $a_{i,j}$ the amount of electricity from plant i that is sent to city j .
- For each demand node j , the plants contribute enough power to meet the demand:

$$a_{1,j}x_1 + a_{2,j}x_2 + \dots + a_{n,j}x_n \geq b_j$$

- If each plant also has a cost c_i , then the total cost is $\sum_{i=1}^n c_i x_i$.

- This example suggests the general linear-programming template:

$$\underbrace{\min_{x \in \mathbb{R}^n} c^t x}_{\text{objective}}$$

$$\underbrace{Ax \leq b, \quad x \geq 0}_{\text{constraints}}$$

- So a linear program chooses the values of $x_1, \dots, x_n$ to optimize a linear objective while satisfying linear constraints. More generally, the objective can be written as either a minimization or a maximization problem.

Definition 1.2.1 An Integer program is an optimization problem over $x \in \mathbb{Z}^n$ that can be written in one of the following six forms:

$$\begin{aligned} \max\{c^T x : Ax \leq b\} \\ \min\{c^T x : Ax \geq b\} \end{aligned}$$

$$\begin{aligned} \max\{c^T x : Ax \leq b, x \geq 0\} \\ \min\{c^T x : Ax \geq b, x \geq 0\} \end{aligned}$$

$$\begin{aligned} \max\{c^T x : Ax = b, x \geq 0\} \\ \min\{c^T x : Ax = b, x \geq 0\} \end{aligned}$$

Definition 1.2.2 A linear program is an optimization problem over $x \in \mathbb{R}^n$ that can be written in one of the following six forms:

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 Remark

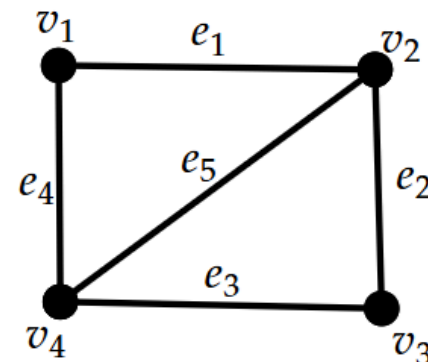
LP's $\in$ P but IP's $\in$ NPH

2. Examples

The *incidence* matrix of a graph G is given by

$$M := M(G) = \begin{matrix} & e_1 & e_2 & \dots & e_m \\ \begin{matrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{matrix} & \begin{bmatrix} & & & \\ & 0/1 & & \\ & & & \\ & & & \end{bmatrix} \end{matrix} \quad M_{ve} := \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases}$$

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$



Each column has $\#1 = 2$

Each row has $\#1 = \deg(v)$

Given G with n vertices and m edges, we seek a vector $\mathbf{x} \in \{0, 1\}^m$ s.t.

- $x_e = 1$ when edge e belongs to the matching, and $x_e = 0$ otherwise.
- maximize $\sum_{e \in E} x_e = \max \mathbf{1}^t \mathbf{x}$.

(We want maximum matching)

This vector $\mathbf{x}$ is the characteristic vector of a matching in G .

So far the above doesn't define a matching!

We need to define constraints.

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We need to define constraints.

- For every vertex $v \in V$: $\sum_{e \in E: v \in e} x_e \leq 1$.

(Every vertex can be matched only once)

2.2 Maximum Matching

Given G with n vertices and m edges, we seek a vector $x \in \{0, 1\}^m$ s.t.

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(Every vertex can be matched only once)

or by matricial form

$$x \in \mathbb{Z}^n$$

$$\max \mathbf{1}^t x$$

$$\begin{matrix} & e_1 & e_2 & \dots & e_m \\
 \begin{matrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & \dots & 0 & 1 \end{bmatrix}
 \end{matrix}$$

M
 $n \times m$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}$$

x
 $m \times 1$

$=$

$$\begin{bmatrix} (Mx)_1 \\ \vdots \\ (Mx)_i \\ \vdots \end{bmatrix}$$

Mx
 $n \times 1$

$$(Mx)_i = \langle v_i, x \rangle$$

#edges incident to v_i
by the given matching

*This is why $Mx \leq 1$
must hold if X is a matching!*

For the max-matching problem, we arrive at the following

IP for maximum matching

$$\max_{x \in \mathbb{Z}^m} \{ \mathbf{1}^t x : Mx \leq \mathbf{1}, x \geq \mathbf{0} \}$$

LP for maximum matching

$$\max_{x \in \mathbb{R}^m} \{ \mathbf{1}^t x : Mx \leq \mathbf{1}, x \geq \mathbf{0} \}$$

This called the:
Corresponding LP Relaxation

Remark

We don't need to bound $x \leq \mathbf{1}$ as if there exists i for which $x_i > 1$. Then, there must exist $j \in [m]$ for which $M_{j,i} < 1$. But, we know that M is the *incidence matrix* of G so that for exactly two indices j_1, j_2 we have $M_{j_1,i} = 1$ and $M_{j_2,i} = 1$.

Given G with n vertices and m edges, we seek a vector $\mathbf{y} \in \{0, 1\}^n$ s.t.

- $y_v = 1$ when vertex v belongs to the **vc**, and $y_v = 0$ otherwise.
- minimize $\sum_{v \in V} y_v = \min \mathbf{1}^t \mathbf{y}$.

(We want minimum vertex-cover)

Constraints: $\forall e \in E : \sum_{v \in e} y_v \geq 1$

(Every edge is covered at least once)

This vector $\mathbf{y}$ is the characteristic vector of a vertex cover in G .

- minimize $\sum_{v \in V} y_v = \min \mathbf{1}^t \mathbf{y}$.

Constraints: $\forall e \in E : \sum_{v \in e} y_v \geq 1$

or by matricial form

$y \in \mathbb{Z}^n$
 $\min \mathbf{1}^t y$

$$\begin{matrix} & v_1 & v_2 & \dots & v_n \\ \begin{matrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix} & = & \begin{bmatrix} (M^t y)_1 \\ \vdots \\ (M^t y)_i \\ \vdots \end{bmatrix}
 \end{matrix}$$

M^t
 $n \times m$

y
 $n \times 1$

My
 $m \times 1$

$(M^t y)_i = \langle e_i, y \rangle$
#vertices covering the edge e_i

This is why $M^t y \geq \mathbf{1}$ must hold if Y is a matching!

For the min-vc problem, we arrive at the following

IP for minimum-vc

$$\min_{\mathbf{y} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{y} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \}$$

LP for minimum-vc

$$\min_{\mathbf{y} \in \mathbb{R}^n} \{ \mathbf{1}^t \mathbf{y} : M \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \}$$

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What if we wanted vc with minimum weight? i.e. every vertex $v \in V$ has weight $w_v \in \mathbb{R}$?

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$$\min_{\mathbf{y} \in \mathbb{Z}^n} \left\{ \underbrace{\mathbf{w}^t \mathbf{y}}_{\sum_{v \in V} w_v \cdot y_v} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \right\}$$

By Konig we know that in bipartite graphs it must hold

IP for maximum matching

$$\max_{\mathbf{x} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{x} : M \mathbf{x} \leq \mathbf{1}, \mathbf{x} \geq \mathbf{0} \}$$

=

IP for minimum-vc

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In fact

LP for maximum matching

$$\max_{\mathbf{x} \in \mathbb{R}^n} \{ \mathbf{1}^t \mathbf{x} : M \mathbf{x} \leq \mathbf{1}, \mathbf{x} \geq \mathbf{0} \}$$

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LP for minimum-vc

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$$\begin{array}{|c|} \hline \text{IP for maximum matching} \\ \max_{\mathbf{x} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{x} : M \mathbf{x} \leq \mathbf{1}, \mathbf{x} \geq \mathbf{0} \} \\ \hline \end{array} = \begin{array}{|c|} \hline \text{IP for minimum-vc} \\ \min_{\mathbf{y} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{y} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \} \\ \hline \end{array}$$

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If G is not bipartite, then in fact

$$\begin{array}{|c|} \hline \text{IP for maximum matching} \\ \max_{\mathbf{x} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{x} : M \mathbf{x} \leq \mathbf{1}, \mathbf{x} \geq \mathbf{0} \} \\ \hline \end{array} \in \mathbf{P} \qquad \begin{array}{|c|} \hline \text{IP for minimum-vc} \\ \min_{\mathbf{y} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{y} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \} \\ \hline \end{array} \in \mathbf{NPC}$$

3. Geometric analysis of LP's

Consider the following **LP**:

$$\max x_1 + x_2$$

s.t.

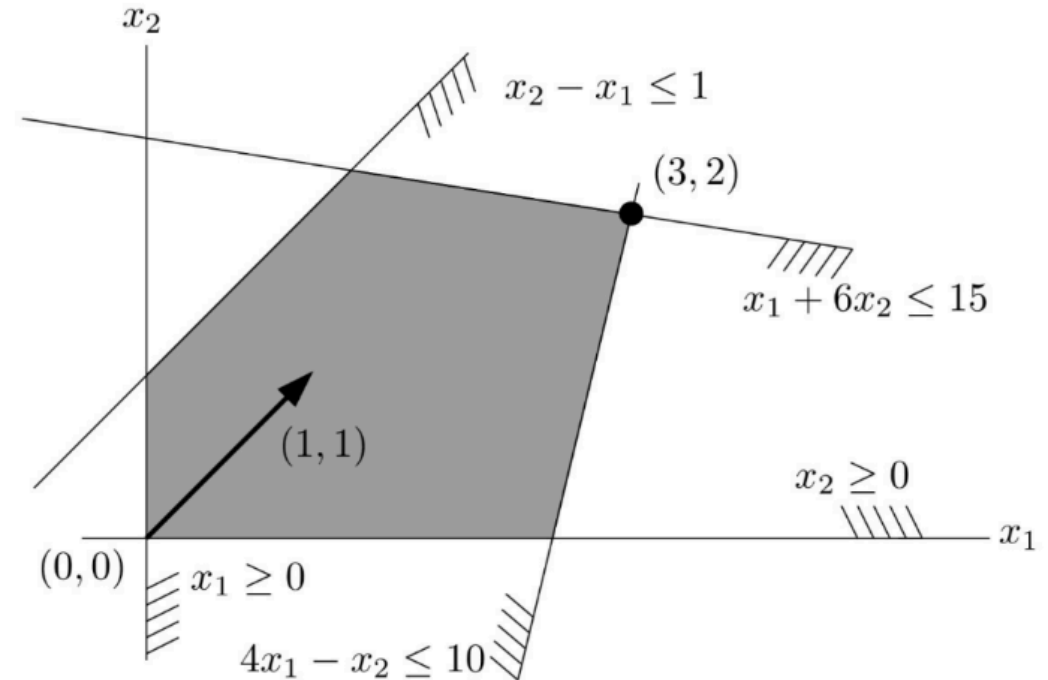
$$-x_1 + x_2 \leq 1$$

$$x_1 + 6x_2 \leq 15$$

$$4x_1 - x_2 \leq 10$$

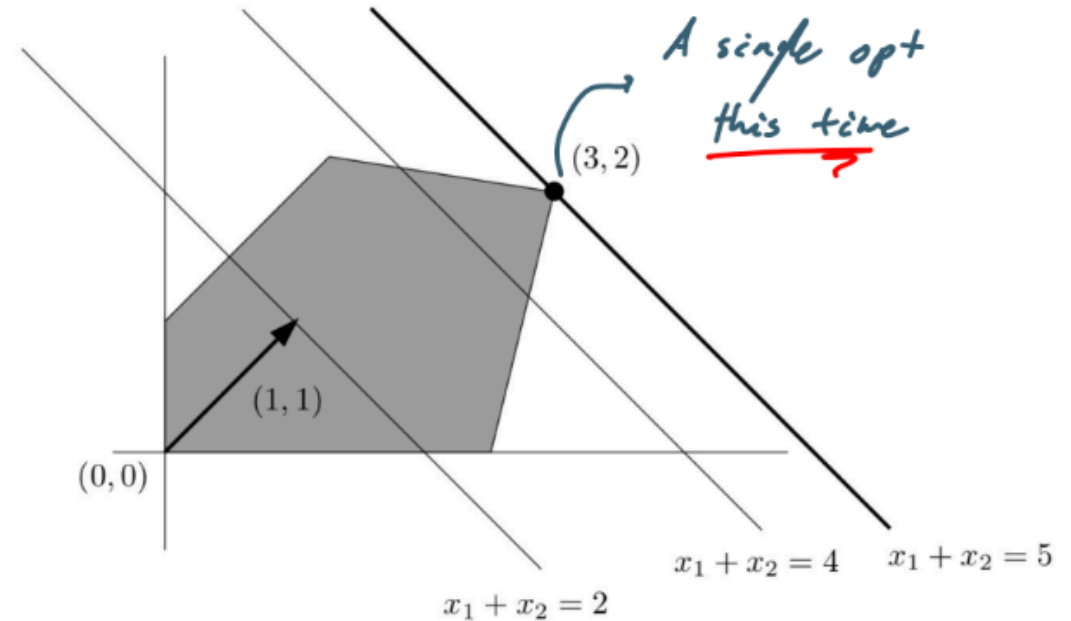
$$x_1 \geq 0$$

$$x_2 \geq 0$$

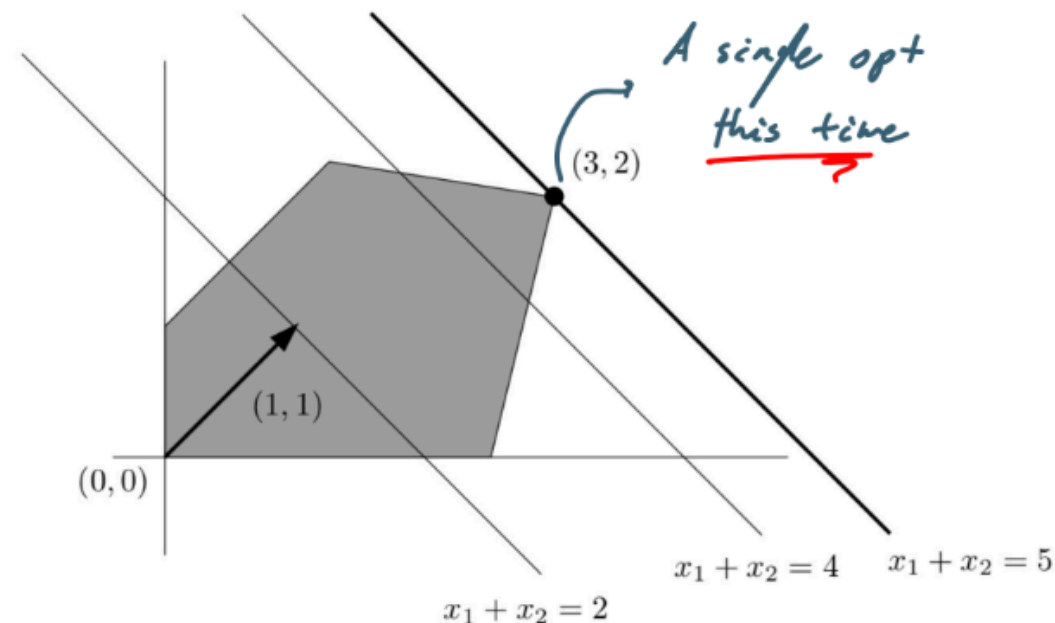
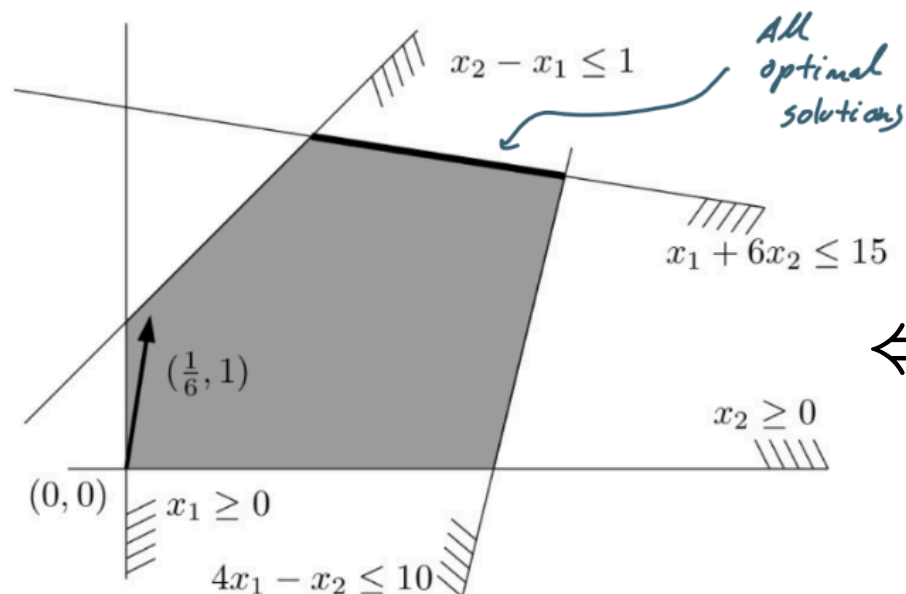


$$\max_{x \in \mathbb{R}^2} (1, 1) \cdot (x_1, x_2) \text{ s.t. } \begin{pmatrix} -1 & 1 \\ 4 & 6 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 \\ 15 \\ 10 \end{pmatrix}, \quad (x_1, x_2) \geq (0, 0)$$

Solving the problem Geometrically, would mean taking the line $x_1 + x_2 = c$ and moving it in the direction $(1, 1)$ until we get out of constraint.



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If we change the constraint to $\max \frac{1}{6}x_1 + x_2$.
Then, we have infinitely many solutions.

Definition 3.2.1 A Linear program in a one of the following 2-forms:

$$\max\{c^T x : Ax = b, x \geq 0\}$$

$$\min\{c^T x : Ax = b, x \geq 0\}$$

is said to be in **equational form**;

Remark

Optimal solution for **LP's** in **equational form** are the easiest to explain and we will focus on those.